Question 1

Define the following terms, within the context of the speech production mechanism:

- Vocal Tract
- Voiced Speech
- Unvoiced Speech
- Pitch Frequency.

1

- **Vocal Tract**

The pharyngeal (mouth) , nasal and oral cavities are usually referred together as Vocal Tract.

Voiced Speech

Voiced sounds are produced when air from the lungs forces the vocal cords to oscillate (vibrate).

Unvoiced Speech

For unvoiced speech, the vocal cords are held open and air flows continuously through them.

There is no vibration of vocal cords as in voiced speech.

- **Pitch Frequency.**
- Pitch is the fundamental frequency of vibration of the vocal cords. i.e 150-200Hz for male and ……..

2

Question 2

- The Vocal Tract is an acoustical tube, which consists of:
  - The lungs, the trachea and the larynx
  - The pharyngeal, oral and nasal cavities
  - The trachea and nasal cavity
  - The oral cavity and larynx
  - None of the above

3

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  - The pharyngeal (mouth), oral and nasal cavities
  - The trachea and nasal cavity
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4

Question 3

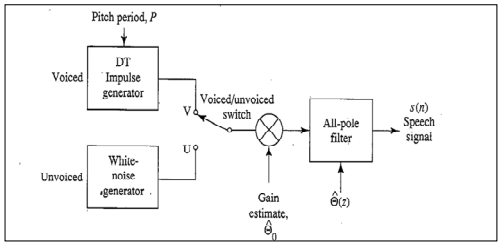
- What are the main characteristics of Voiced and Unvoiced Speech:
  - in the time domain, and
  - in the frequency domain.

- Slide 4 part 2

5

Question 3a

Provide the general block diagram of a Speech Synthesis Process that is based on the "Binary Excitation Model".



6

Question 4

- **The Pitch frequency is:**
  - The frequency of oscillation of the vocal tract
  - The frequency at which the vocal tract resonates
  - The dominant frequency in the speech signal
  - The frequency of oscillation of the vocal cords
  - None of the above
- **Different vocal tract (VT) configurations (shapes) result to:**
  - Different pitch frequencies
  - Different speech sounds
  - Different modes of speech production
  - Different energies in the speech signal
  - All of the above
-

7

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8

Question 5

- What does the **general** sampling (Nyquist) theorem state?
- Homework

9

Question 6

- What are the main aims of speech coding systems?

10

Answer

- The general aim of a Speech Coding process is to digitize the input signal using **the least number of bits per second** (minimum bit rate).
- Compression is achieved by exploiting the intrinsic properties of the signal.
- Thus speech coding algorithms **seek to remove redundant information** that exists in the speech and which, when transmitted, increases unnecessarily output bit rates.

11

Question 7

- Mention the broad types of speech coding systems and their bit rate range of operation.

12

- **Waveform Coders** :
  - bit rates of 16 Kb/sec and higher.
  - Time Domain (PCM, DPCM, ADPCM, ADM, APC, Sub-band Coding)
  - Frequency Domain (ATC, Harmonic, Coding)
-
- **Vocoders :**
  - low bit rates, usually 2.4 Kb/sec
  - (Channel, Formant, Homomorphic, LPC, Phase)
-
- **Hybrid Systems** : Low bit rates, 8 to 1.2 Kb/sec
  - (CELP, MPE-LPC, Sinusoidal Coding, Multiband Excitation)
-

13

Question 8

Vocoders can operate at:

a) Very low bit rates with communications quality speech
b) High bit rates with network quality speech
c) Very low bit rates with synthetic quality speech
d) High or low bit rates with good or synthetic quality speech, respectively
e) None of the above

14